[METRIC]
A-A-2263E
February 4, 1997
SUPERSEDING
A-A-2263D
August 20, 1986

# COMMERCIAL ITEM DESCRIPTION

## CALCULATOR, ELECTRONIC

## (DESK PORTABLE, 10 DIGITS, 1 OR MORE MEMORIES) (PRINT/DISPLAY)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE:

This description covers a general purpose, desk portable type electronic calculator intended for general office use.

2. SALIENT CHARACTERISTICS:

2.1 Dimensions (minimum): Not more than 171.5 mm (6-3/4 inches) by 241.3 mm (9-1/2 inches) nor less than 114.3 mm (4-1/2 inches) by 177.8 mm (7 inches). Dimensions apply to the calculator body excluding paper holder brackets or cord ferrules.

2.2 Power: Battery powered with adapter for 110-120 volt, AC, 60 Hz. The calculator must carry the approval of a nationally recognized testing laboratory recognized by the Occupational Safety and Health Administration under Appendix A to 29 CFR 1910.7.

2.3 Convenience supplies: Each calculator must have an operating manual that contains instructions for each function key.

2.4 Workmanship: Calculators shall be free from faulty construction or any characteristic or defect which may affect appearance, usability, serviceability, durability or which may constitute a hazard to the user.

| Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Engineering Branch, New York, NY 10278. |
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2.5 General Features:

2.5.1 Display capacity (minimum) - 10 digits.

2.5.2 Display digit height (minimum) - 6.4 mm (1/4 inch).

2.5.3 Negative sign for negative entries and answers.

2.5.4 Clear capabilities - shall, at a minimum, have a clear entry key, which upon depression clears the last keyboard entry, and a separate clear key which upon depression clears the contents of the working register.

2.5.5 Constant for multiplication and division: Allows multiplication or division of a series of numbers by one number, without reentry of the number.

2.5.6 Repeat addition and subtraction: Allows repeat addition or subtraction of a number by mere depression of a key such as the plus or minus key, without reentry of the number.

2.5.7 Separate keys for totals and subtotals.

2.5.8 Floating decimal in entries and answers.

2.5.9 Capability to pre-select a minimum of 3 fixed decimal positions.

2.5.10 Add Mode: Allows entry of numbers that contain decimals without the need to enter the decimal point. At a minimum, must have the position of being fixed for 2 decimal places.

2.5.11 Percent Keys: A minimum of 2 keys including a Mark Up key. The mark up is a percentage of the selling price (not the cost price). If an item cost $100, and we want to mark it up by 20%, the selling price would be $125, and not $120. If X = cost price, Y = price after mark up, and MU = mark up (which is a percentage of Y):

$$Y = X + MU.Y$$
$$Y - MU.Y = X$$
$$Y(1-MU) = X$$
$$Y = X / (1 - MU) \qquad (1)$$

Thus if x = 100 and MU =20%

$$Y = 100 / 1 - 20\% = 100 / 1 - 0.20 = 100 / 0.80 = 125$$

Equation (1) is what the mark up key calculates.



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2.5.12 Rounding: Minimum of 5/4 round off which allows the last decimal digit displayed in an answer to be increased by 1, if the following digit would have been a 5 or greater.

2.5.13 Memory: A minimum of one memory with the ability of storing a number for later recall and reuse. It shall have a minimum of 4 keys ; one for entering a number "positively" into memory, one for entering a number "negatively" into memory, one for reading contents of memory, and one for clearing memory.

2.5.14 Roll-over Indexing: Allows depression of a key, before releasing the previous key.

2.5.15 Raised key tops: Keys shall protrude a minimum of 3.175 mm (1/8") from surface of calculator.

2.6 Printer Features:

2.6.1 Printer speed: A minimum of 2.5 lines per second. This can be measured by indexing a "1" and operating the repeat addition key for 15 seconds. A total is then taken and divided by 15 to find the number of lines per second.

2.6.2 Digit capacity - 10 (minimum).

2.6.3 Digit height (minimum) - 2.4 mm (3/32 inch)

2.6.4 Paper type - regular plain paper, non-thermal imaging.

2.6.5 Paper width - must accommodate 57.2 mm (2-1/4 inches).

2.6.6 Item counter: When activated, the number of entries are counted.

2.6.7 Non-add key: Used for printing numbers that do not affect the calculations, such as notations and dates.

2.6.9 Subtotal print from display to printer: If calculations are performed with only the display on, and subsequently the printer is switched on, a subtotal of those calculations performed while the display alone was on can be printed on the tape.



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## 3. REGULATORY REQUIREMENTS:

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

## 4. QUALITY ASSURANCE PROVISIONS:

Product Conformance: The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards and quality assurance practices and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

## 5. PACKAGING:

Packaging, packing, labeling and marking shall be as specified in the contract or purchase order.

## 6. NOTES:

6.1 Ordering Data: Purchasers when ordering should refer to this commercial item description and specify the quantity needed.

6.2 Application for copies of documents referenced herein: The issue date in effect on date of invitation for bids or request for proposal shall apply.

6.2.1 Federal Acquisition Regulation (FAR), and Code of Federal Regulations (CFR) are available from the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

6.3 NSN List:
7420-01-174-5221

Military Coordinating Activity
Air Force 99

Preparing Activity
GSA-FSS

Military Reviewing Activity
Air Force 84



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